



Project 1

Josh Ginnett and Zach Santiana



System Level Design Choices

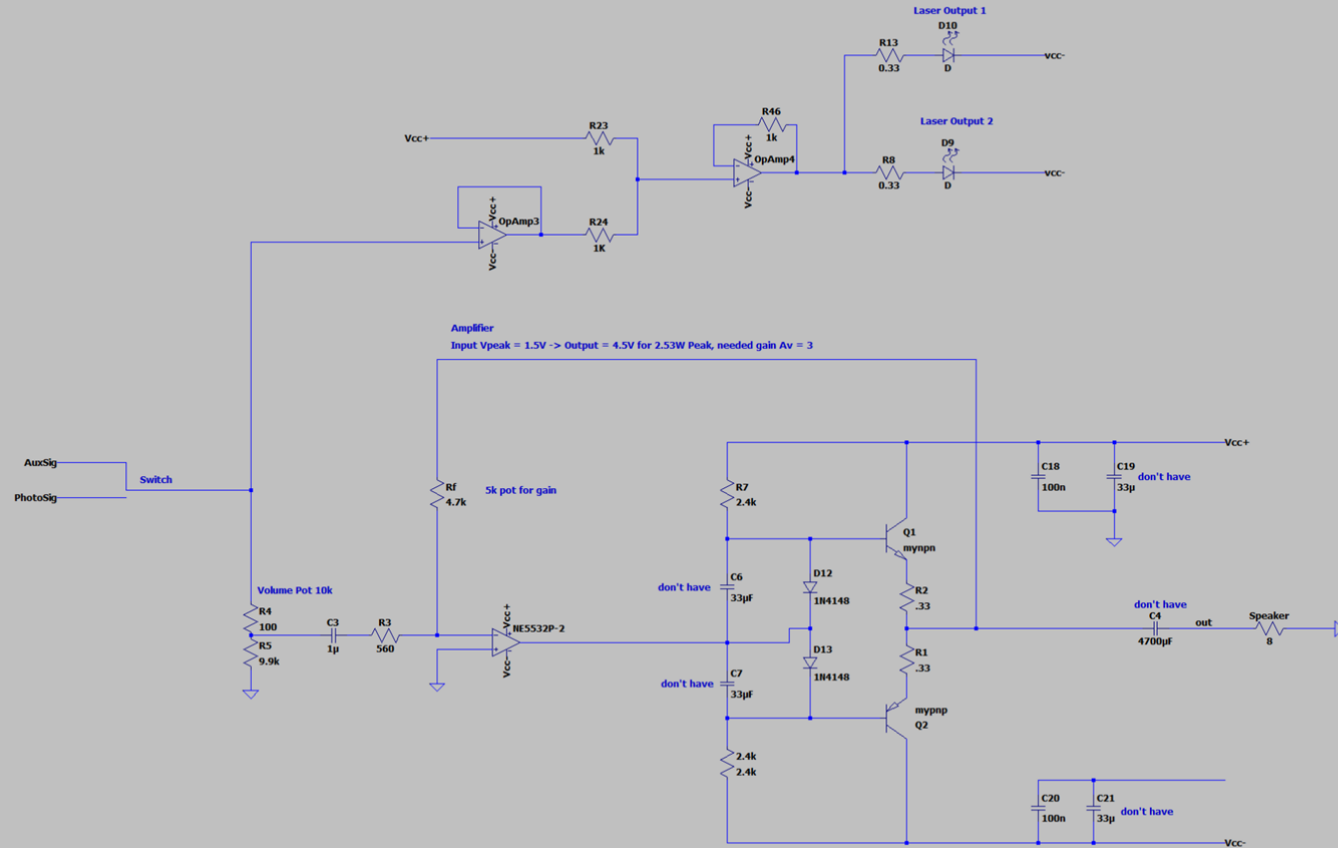
Selectable Input Source: A dedicated switch allows the user to choose between aux input or the solar-cell photodetector as the signal source. This enables both direct wired playback and optical signal transmission from another student's laser setup.

Adjustable Gain and Volume: Independent controls for amplifier gain (can be used to boost volume or add distortion if desired) and output volume.

Integrated Power System: Powered by a 9V battery with a built-in power switch and an LED status indicator, allowing the battery to remain connected without draining while providing clear visual feedback when the device is active.

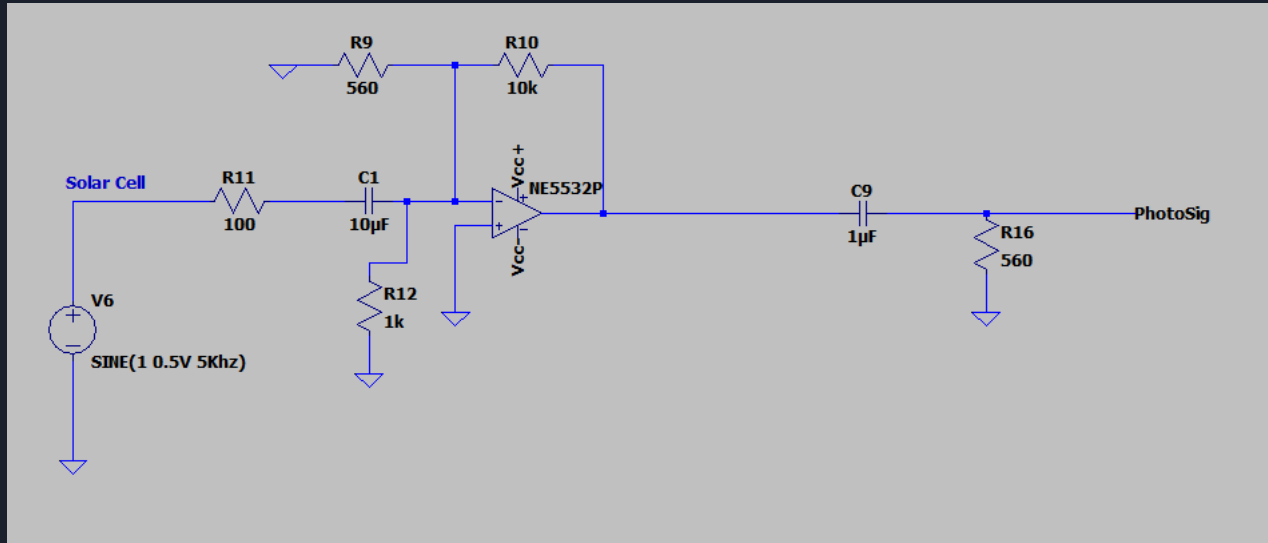
Dual Laser Outputs With Dedicated Amplifier Paths: Two laser transmitters, each driven by its own amplifier stage, allow optical transmission of the audio signal with minimal added gain or noise. Using separate amplifier paths isolates the transmit channels and ensures cleaner modulation onto the laser diodes.

Circuit Design Overview



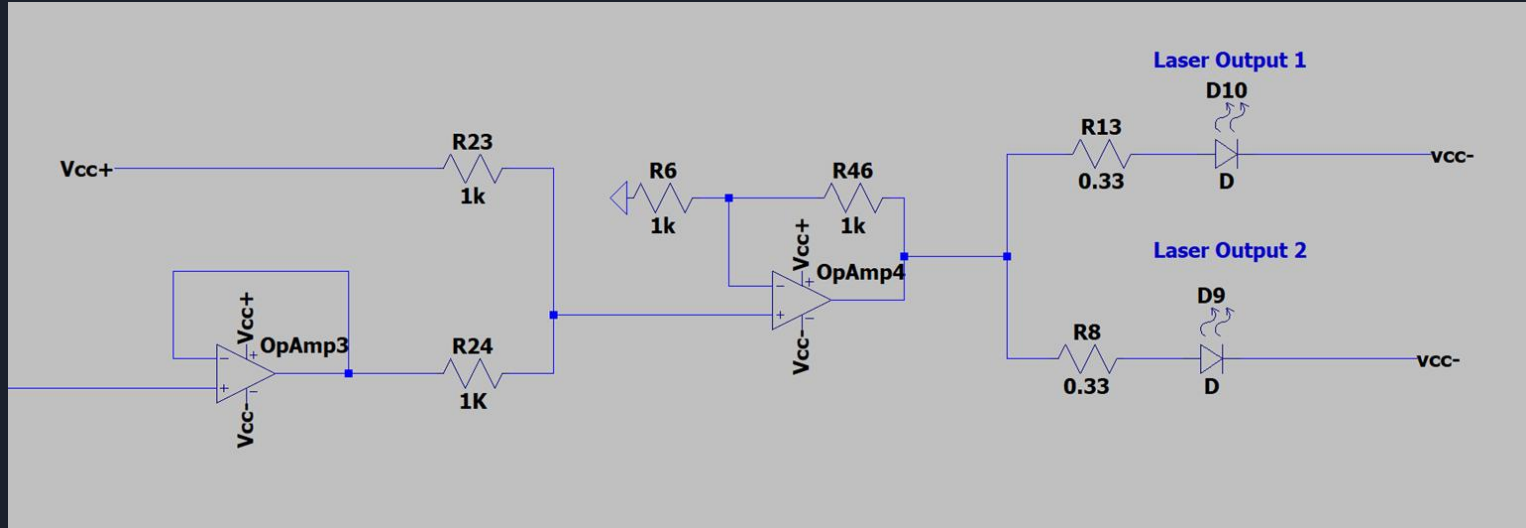
Preamp for Solar Cell

Measurements showed the aux input at approximately 0.989 Vrms, while the solar-cell output was only about 0.353 Vrms. Because the optical signal is significantly weaker, a preamplifier was required to raise the solar-cell signal to a comparable level before feeding it into the main amplifier. A high pass filter is implemented after pre amplifying to reduce signal noise from the solar cell.



Laser Output Stage

The selected input signal is first buffered with a voltage follower for isolation, then passed into a non-inverting summing amplifier to introduce the required DC bias for laser modulation.



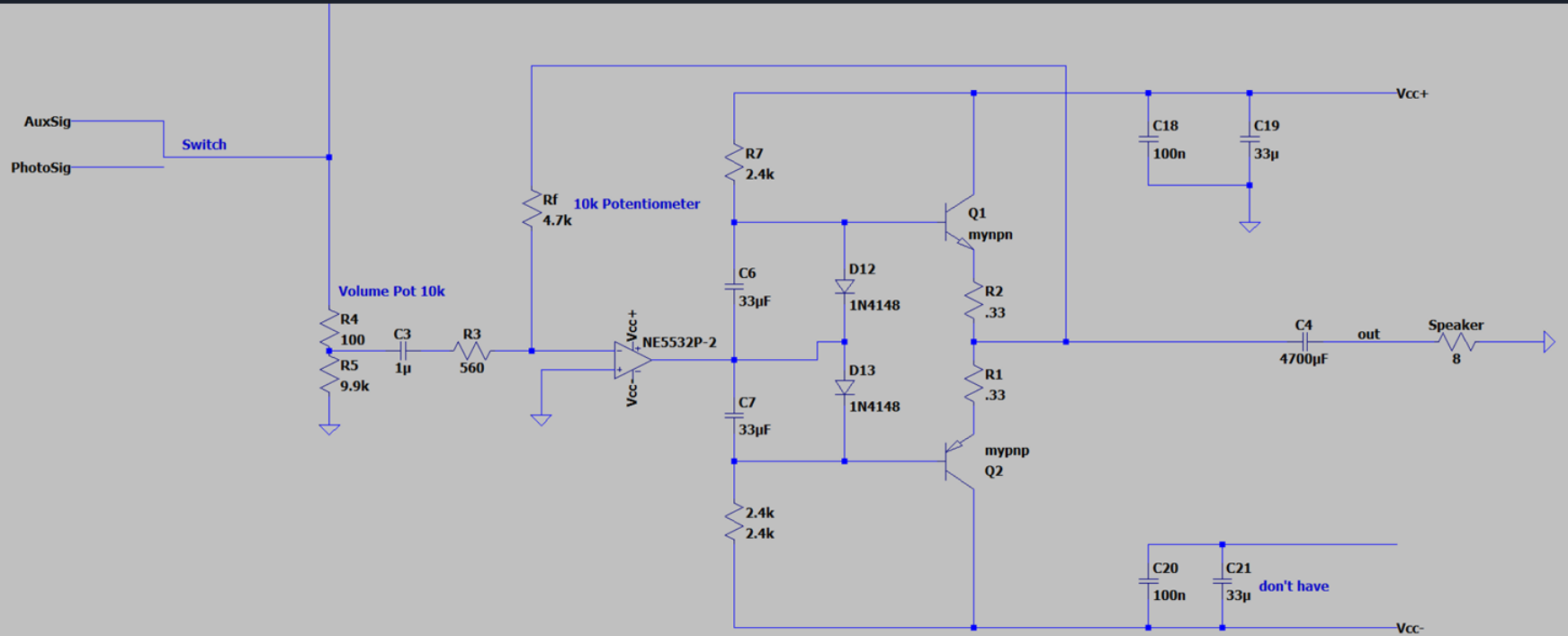


Main Amplifier

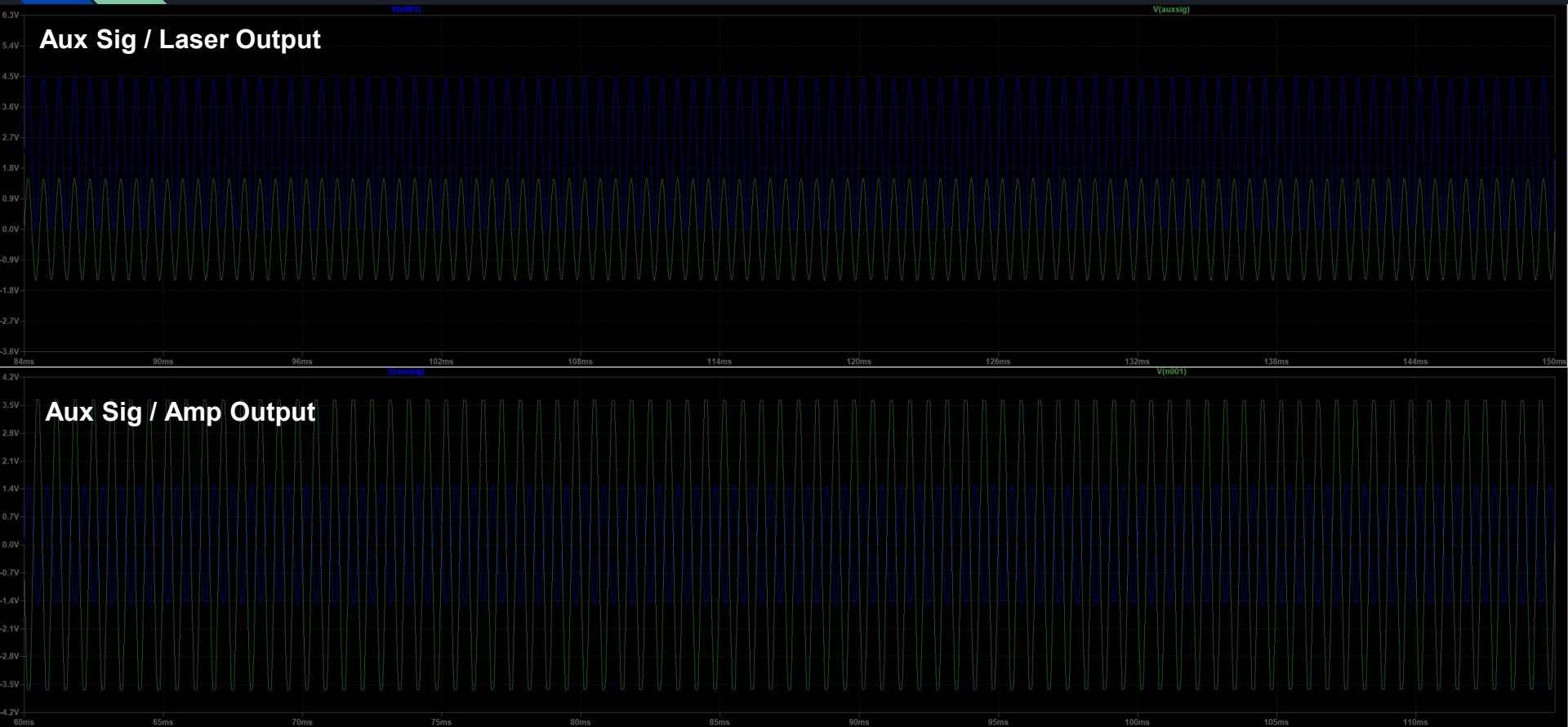
The amplifier uses an NE5532P op-amp configured in an inverting topology, with a 10 k Ω potentiometer in the feedback path to provide adjustable gain. Although the NE5532P can source up to about 70 mA, this is not sufficient to drive the speaker at the required power level. To supply the needed current, a diode-biased push-pull output stage is added, providing unity voltage gain but substantial current gain. The 1N4148 diodes bias the BD139/BD140 transistor pair, reducing crossover distortion while keeping the quiescent current low (~ 0.2 mA) for improved efficiency.

The main amplifier is fed by the input selector switch and a 10 k Ω volume potentiometer connected to ground. This allows the user to control volume by bleeding part of the signal to ground when the potentiometer is set to lower resistance.

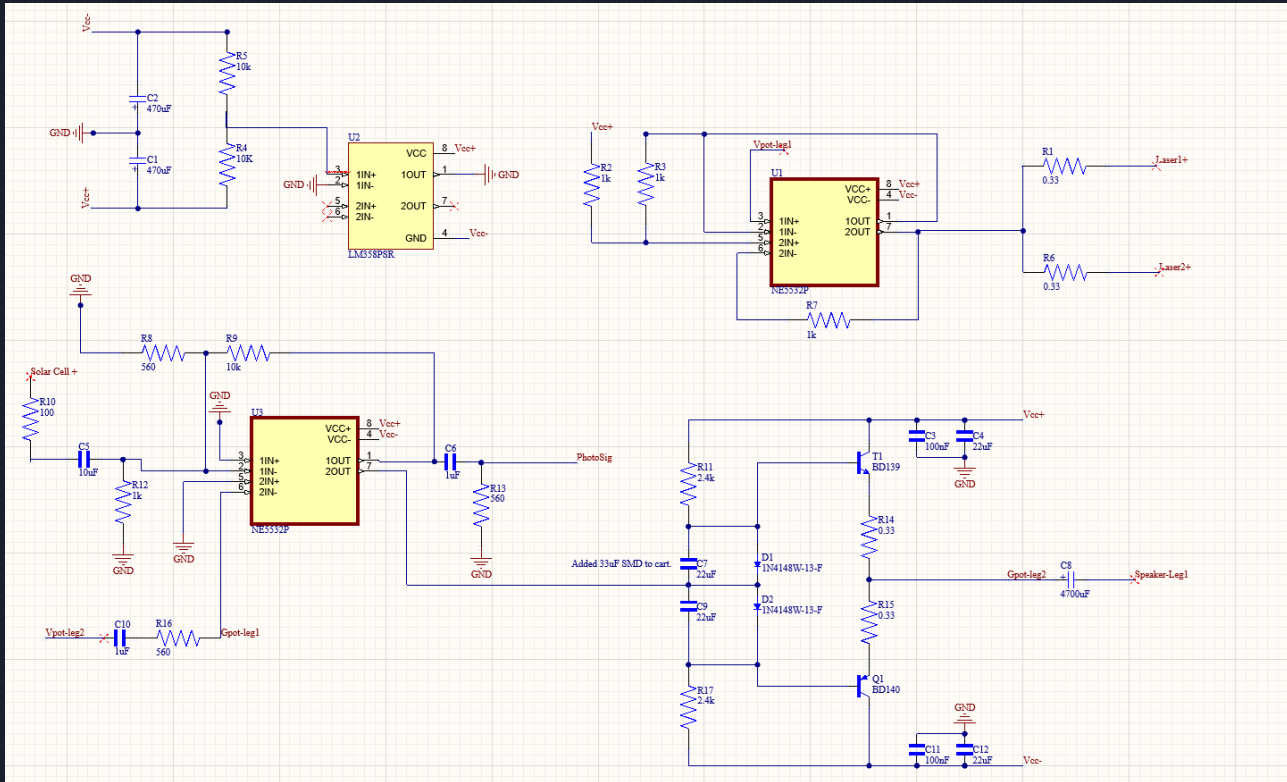
Main Amplifier



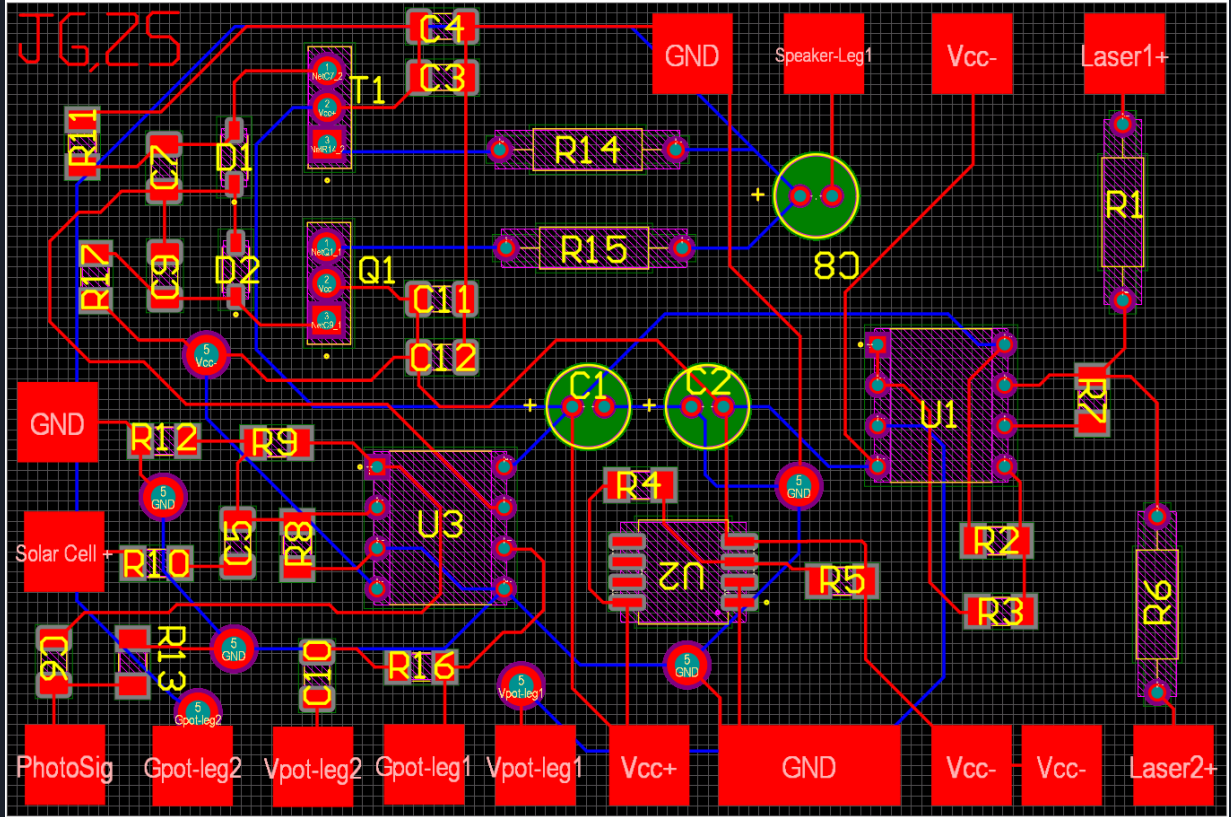
LT Spice Simulations



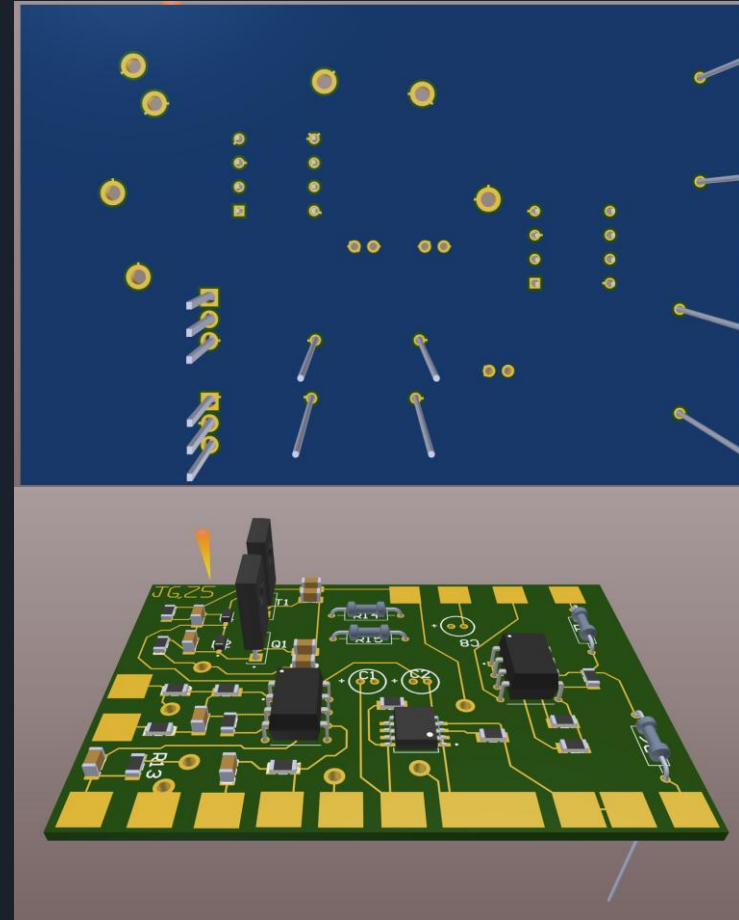
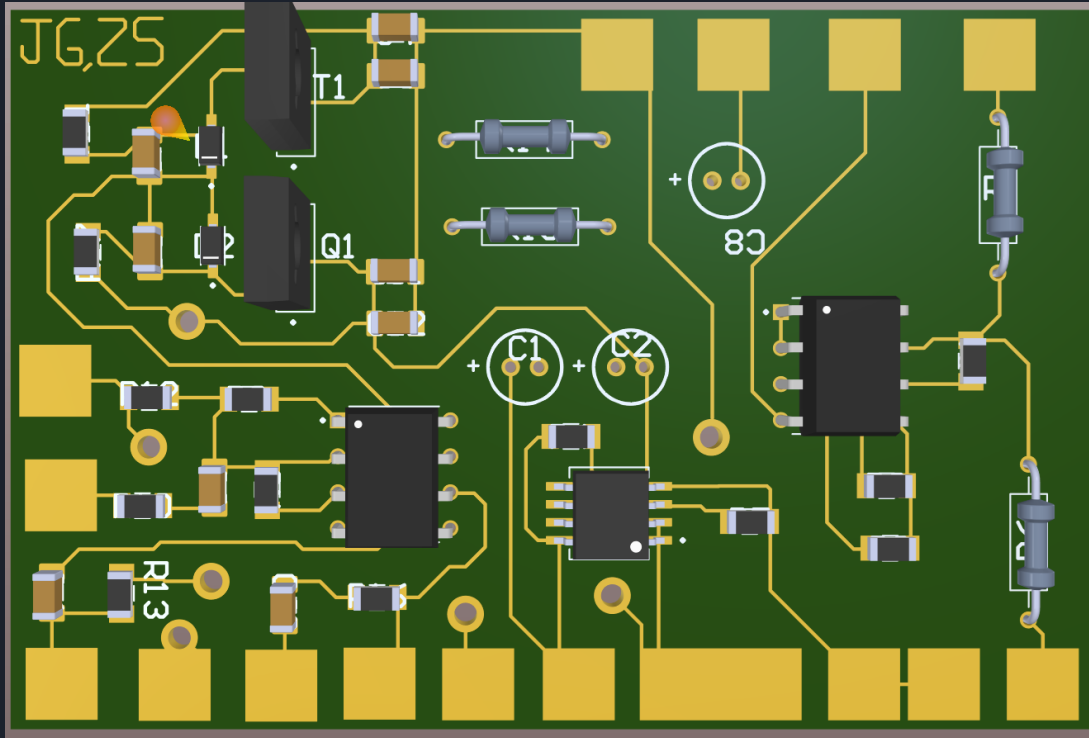
Altium Schematic



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3D View



Assembled Project



Video Demonstration

<https://www.youtube.com/watch?v=YMSiEr61HQ0>

